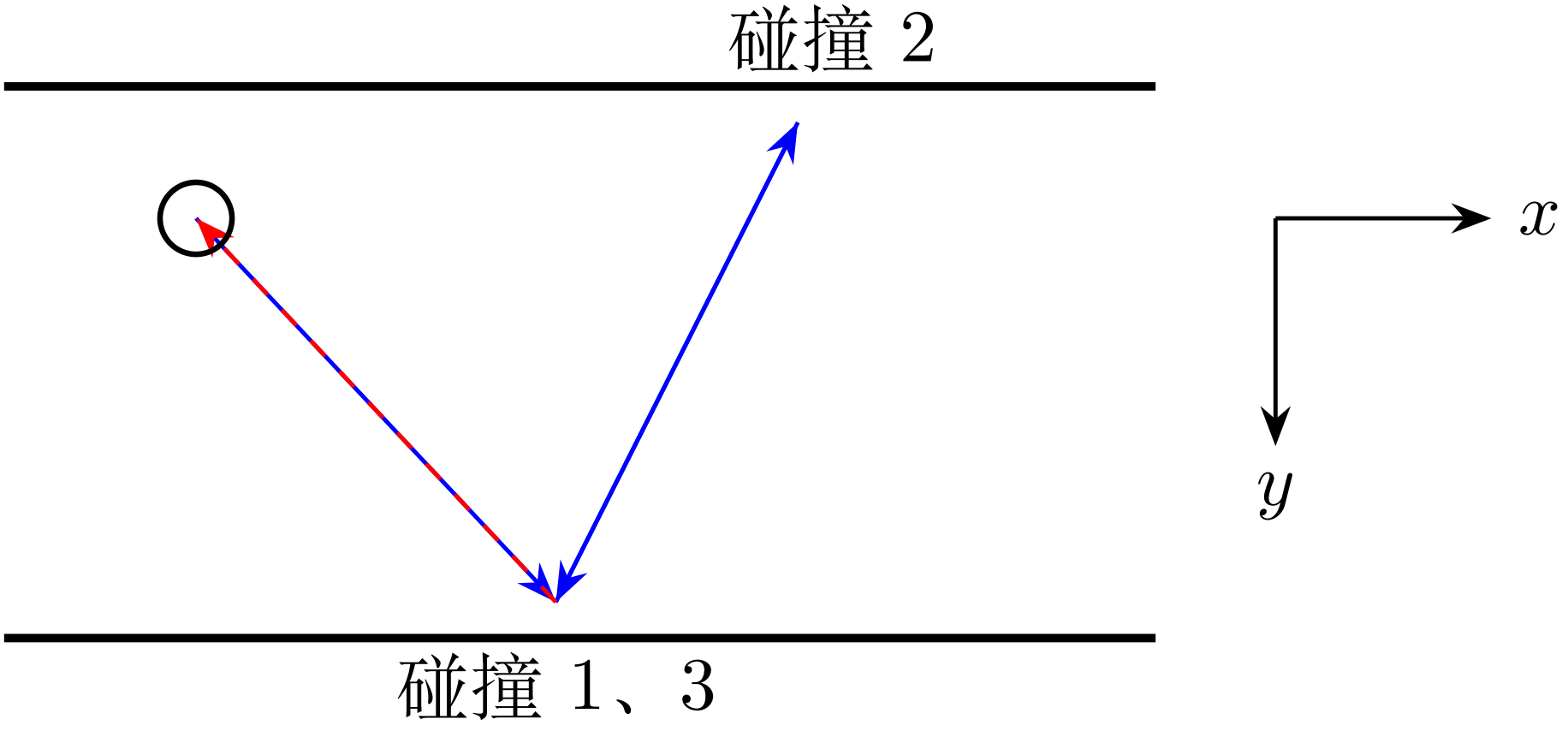示意取 $r\omega_0/v_0 = 5/26$，三碰后反向